Generalized linear models

Devianz

Prof. Dr. Andreas Groll

Winter semester 20/21

Deviance - Contents

Deviance

General problems

- Goal: Measure for the discrepancy between the data and the model fit
- Usage of similar structure as for the log-likelihood ratio statistic, i.e. as for

$$\lambda = -2\log(\mathrm{LR}) = -2\left(l(\tilde{\beta}) - l(\hat{\beta})\right)$$

- However, in this case we consider
  - $l(\mathbf{y}, \hat{\mu})$ log-likelihood of observed data $(y_1, \dots, y_n)$ and the estimated means $\hat{\mu} = (\hat{\mu}_1, \dots, \hat{\mu}_n)$, $\hat{\mu}_i = h(\underline{x}_i^T \hat{\underline{\beta}})$
  - $l(\mathbf{y}, \hat{\mathbf{y}})$ —— " —— and the best possible fit $\hat{y}$: $\hat{\underline{y}} = \underline{y}$ ungr. data, $\hat{\underline{y}} = \underline{y}$ gr. data

Deviance

$h(\underline{x}^T \beta)$

obs.

Deviance

For GLMs, the deviance is defined as:

obs. or group means

$$D(\mathbf{y}, \hat{\boldsymbol{\mu}}) = -2 \cdot \phi \left(l(\mathbf{y}, \hat{\boldsymbol{\mu}}) - l(\mathbf{y}, \hat{\mathbf{y}}) \right)$$

Scaled deviance

$$D(\mathbf{y}, \hat{\boldsymbol{\mu}})^{+} = \frac{D(\mathbf{y}, \hat{\boldsymbol{\mu}})}{\phi}$$

$\hat{=}$ LR-statistic in the saturated model ($\hat{=}\ p = n$)

The deviance is maximized by the ML estimator, the second part $l(\mathbf{y}, \hat{\mathbf{y}})$ is constant.

$\hat{\mu} = h(\underline{x}^T \hat{\beta}_{ML})$

Deviance

In a GLM (because of the exponential family) we have:

$$l(\mathbf{y}, \hat{\boldsymbol{\mu}}) = \frac{1}{\phi} \sum_{i=1}^{n} (y_i \vartheta_i - b(\vartheta_i)) + \sum_{i=1}^{n} c(y_i, \phi)$$

$$\rightarrow D(\mathbf{y}, \hat{\boldsymbol{\mu}}) = -2\cancel{\phi} \sum_{i=1}^{n} \left(\left[\frac{y_i \vartheta(\hat{\mu}_i) - b(\vartheta(\hat{\mu}_i))}{\cancel{\phi}} + \cancel{c(y_i, \phi)} - \frac{y_i \vartheta(y_i) - b(\vartheta(y_i))}{\cancel{\phi}} - \cancel{c(y_i, \phi)} \right] \right)$$

$$\Rightarrow 2 \sum_{i=1}^{n} \Big(y_i \big(\vartheta(y_i) - \vartheta(\hat{\mu}_i) \big) - \big(b(\vartheta(y_i)) - b(\vartheta(\hat{\mu}_i)) \big) \Big)$$

Deviance

Examples:

Normal distribution: $\vartheta(\mu_i) = \mu_i, \quad b(\vartheta(\mu_i)) = \frac{\mu_i^2}{2}$

$$\to D(\mathbf{y}, \hat{\boldsymbol{\mu}}) = 2 \sum_{i=1}^{n} \left[y_i (y_i - \hat{\mu}_i) - \left(\frac{y_i^2}{2} - \frac{\hat{\mu}_i^2}{2} \right) \right]$$

$$= 2 \sum_{i=1}^{n} \left[y_i^2 - y_i \hat{\mu}_i - \frac{y_i^2}{2} + \frac{\hat{\mu}_i^2}{2} \right]$$

$$= \sum_{i=1}^{n} \left(y_i^2 - 2 y_i \hat{\mu}_i + \hat{\mu}_i^2 \right) = \sum_{i=1}^{n} (y_i - \hat{\mu}_i)^2$$

$\Rightarrow$ Deviance corresponds to Squared Error !

Deviance

$$y_i \in \{0,1\}$$

Bernoulli distribution: $\vartheta(\mu_i) = \log(\frac{\mu_i}{1-\mu_i})$,
$b(\vartheta(\mu_i)) = -\log(1-\mu_i)$

$$\to D(\mathbf{y}, \hat{\boldsymbol{\mu}}) = 2\sum_{i=1}^{n} \left(y_i \left(\log\left(\frac{y_i}{1-y_i}\right) - \log\left(\frac{\hat\mu_i}{1-\hat\mu_i}\right)\right) - \left(-\log(1-y_i) + \log(1-\hat\mu_i)\right)\right)$$

$$= 2\sum_{i=1}^{n} \left[y_i \left(\log\left(\frac{y_i}{1-y_i} \cdot \frac{1-\hat\mu_i}{\hat\mu_i}\right)\right) - \log\left(\frac{1-\hat\mu_i}{1-y_i}\right)\right]$$

$$= 2\sum_{i=1}^{n} \left[y_i \log\left(\frac{y_i}{\hat\mu_i}\right) + y_i \log\left(\frac{1-\hat\mu_i}{1-y_i}\right) - \log\left(\frac{1-\hat\mu_i}{1-y_i}\right)\right]$$

$$= 2\sum_{i=1}^{n} \left[y_i \log\left(\frac{y_i}{\hat\mu_i}\right) + (1-y_i) \log\left(\frac{1-y_i}{1-\hat\mu_i}\right)\right]$$

Deviance

Problem for binary response variable: y_i can't be inserted directly due to "$\log(0)$".
Hence, initially regard individual terms:

$$L(\pi_i) = \pi_i^{y_i} \cdot (1-\pi_i)^{1-y_i}$$

$$l(y_i, \hat{\mu}_i) = y_i \log(\hat{\mu}_i) + (1-y_i)\log(1-\hat{\mu}_i)$$

$$l(y_i, \hat{y}_i) = y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)$$

First regard $l(y_i, \hat{y}_i)$

$$y_i = 0: \quad l(y_i, \hat{y}_i) = \overbrace{0 \cdot \log(0)}^{:= 0} + 1 \cdot \log(1) = 0$$

$$y_i = 1: \quad l(y_i, \hat{y}_i) := 1 \cdot \log(1) + \underbrace{0 \cdot \log(0)}_{= 0} = 0$$

Deviance

$\rightarrow l(y_i, \hat{y}_i) = 0 \quad \forall y_i \in \{0, 1\}$

$\rightarrow D(\mathbf{y}, \hat{\boldsymbol{\mu}}) = -2 \, l(y_i, \hat{\mu}_i)$

$$= -2 \sum_{i=1}^{n} \underbrace{\left(y_i \log(\hat{\mu}_i) + (1-y_i) \log(1-\hat{\mu}_i) \right)}$$

$$= \begin{cases} \log(\hat{\mu}_i), & \text{if } y_i = 1 \\ \log(1-\hat{\mu}_i), & \text{if } y_i = 0 \end{cases}$$

$$= -2 \sum_{i=1}^{n} \log\left(1 - \underbrace{|y_i - \hat{\mu}_i|}\right)$$

"Difference between data & fit"

Deviance

Does the deviance possess an (asymptotic) distribution?

- For non-grouped data, in general no!
- Example: Normal distribution

$$D(\mathbf{y}, \hat{\boldsymbol{\mu}}) = \sum_{i=1}^{n} (y_i - \hat{\mu}_i)^2 \to \infty$$

$n \to \infty$

no reasonable asymptotic distribution

As Goodness-of-fit measure, the deviance therefore is impractical for non-grouped data; it can only used sporadically as informal measure for comparing two link functions.

deviance(link a) < deviance (link b)

$\Rightarrow$ a better fit with link function a

Deviance for grouped data

- Situation as in Section 3.2.3 about grouped data:
  $y_{ij}, \qquad j = 1, \ldots, n_i, i = 1, \ldots, G$ individual observations
  $\bar{y}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} y_{ij}$ group mean of group i
- Log-likelihood:

$$l(\hat{\vartheta}) = \sum_{i=1}^{G} \frac{\bar{y}_i \vartheta_i - b(\vartheta_i)}{\phi / n_i} + \tilde{c}(\bar{y}_i, \phi / n_i)$$

- Deviance:

$$\begin{aligned} D(\bar{\mathbf{y}}, \hat{\boldsymbol{\mu}}) &= -2 \cdot \phi \cdot (l(\bar{\mathbf{y}}, \hat{\boldsymbol{\mu}}) - l(\bar{\mathbf{y}}, \bar{\mathbf{y}})) \\ &= 2 \cdot \sum_{i=1}^{G} n_i \Big[\bar{y}_i \left(\vartheta(\bar{y}_i) - \vartheta(\hat{\mu}_i) \right) \\ &\qquad - b(\vartheta(\bar{y}_i)) + b(\vartheta(\hat{\mu}_i)) \Big] \end{aligned}$$

Deviance for grouped data

\# parameters
(covariates + intercept)

Asymptotics of deviance for grouped data

$$D(\bar{\mathbf{y}}, \hat{\boldsymbol{\mu}}) \overset{a}{\sim} \chi^2_{G-\overbrace{p-1}}$$

\# groups

Thus, deviance can be used as goodness-of-fit measure or goodness-of-fit test statistic for grouped data.

Deviance for grouped data

$y_{ij} \in \{0, 1\}$ $\quad \bar{y}_i \in [0; 1]$

Example Bernoulli / binomial distribution

$y_{ij} \sim Ber(\pi_i) \Rightarrow n_i \bar{y}_i \sim B(n_i, \pi_i)$

$$\Rightarrow D(\bar{y}_i, \hat{\mu}_i) = -2\left(l(\bar{y}_i, \hat{\mu}_i) - l(\bar{y}_i, \bar{y}_i)\right)$$

$$= -2 \sum_{i=1}^{G} \left[n_i \bar{y}_i \log(\hat{\mu}_i) + n_i(1-\bar{y}_i)\log(1-\hat{\mu}_i) - n_i \bar{y}_i \log(\bar{y}_i) + n_i(1-\bar{y}_i)\log(1-\bar{y}_i)\right]$$

$$= 2 \sum_{i=1}^{G} n_i \left[\bar{y}_i \log\left(\frac{\bar{y}_i}{\hat{\mu}_i}\right) + (1-\bar{y}_i)\log\left(\frac{1-\bar{y}_i}{1-\hat{\mu}_i}\right)\right]$$

Deviance for grouped data

Pearson Statistic

Alternative test statistic for grouped data:

$$\chi^2 = \sum_{i=1}^{G} \frac{(y_i - \hat{\mu}_i)^2}{v(\hat{\mu}_i)/n_i}$$

Deviance for grouped data

Asymptotics Pearson statistic

Generally for all GLMs:

$$\chi^2 = \sum_{i=1}^{G} \frac{(y_i - \hat{\mu}_i)^2}{v(\hat{\mu}_i)/n_i} \overset{a}{\sim} \phi\chi^2_{G-p-1}$$

For unknown ϕ, replace it by the ML estimator and increase p by one.

Deviance analysis

Consider hierarchical models:

basic model

full

$$\widetilde{M} \subset M$$

submodel

e.g.:

$$\widetilde{M} : g(\mu) = \beta_0$$

$$M : g(\mu) = \beta_0 + \mathbf{x}^\top \boldsymbol{\beta}$$

In the following:

M : GLM

$\widetilde{M}$: GLM with linear restriction

$\underline{\underline{C}}\, \beta = \underline{d}$

$\operatorname{rank}(\underline{\underline{C}}) = r$

Deviance analysis

Goal of the deviance analysis:

- Quantify the difference in the goodness-of-fit of M and $\widetilde{M}$ with regard to given data
- Determine which model is more suitable

$\longrightarrow$ e.g. via statistical tests

Deviance analysis

- $\hat{\boldsymbol{\mu}}$ is estimator under M
- $\tilde{\boldsymbol{\mu}}$ is estimator under $\widetilde{M}$

$\underline{\hat{y}} = y$ individual data case

Deviance of model M:

$$D(M) = -2\phi\,(l(\mathbf{y}, \hat{\boldsymbol{\mu}}) - l(\mathbf{y}, \mathbf{y}))$$

Deviance of model $\widetilde{M}$:

$$D(\widetilde{M}) = -2\phi\,(l(\mathbf{y}, \tilde{\boldsymbol{\mu}}) - l(\mathbf{y}, \mathbf{y}))$$

The difference: (conditional deviance)

$$D(\widetilde{M} \mid M) = D(\widetilde{M}) - D(M) = -2\phi\,(l(\mathbf{y}, \tilde{\boldsymbol{\mu}}) - l(\mathbf{y}, \hat{\boldsymbol{\mu}}))$$

Deviance analysis

This corresponds to $= -2 \log \frac{L(\underline{y}, \tilde{\underline{\mu}})}{L(\underline{y}, \hat{\underline{\mu}})} \hat{=} -2 \log (LR)$

$$\frac{D(\widetilde{M} \mid M)}{\phi} = -2\left(l(\mathbf{y}, \tilde{\boldsymbol{\mu}}) - l(\mathbf{y}, \hat{\boldsymbol{\mu}})\right) ,$$

the log-likelihood ratio test statistic $\widetilde{M}$ given M (i.e. $\widetilde{M}$ corresponds to H_0).

Difference of deviances is LR

$$\frac{D(\widetilde{M} \mid M)}{\phi} \overset{a}{\sim} \chi^2_r$$

Deviance analysis

Deviance analysis

The deviance analysis uses the following partition:

$$D(\widetilde{M}) \;=\; D(\widetilde{M} \mid M) \;+\; D(M)$$

discrepance of data $\Leftrightarrow \tilde{M}$

increase of deviance when changing from $M \Rightarrow \tilde{M}$

discrepance of data $\Leftrightarrow M$

Deviance analysis

Example: Normal distribution, linear model:

$$\text{SSE}(\widetilde{M}) \;=\; \text{SSE}(\widetilde{M} \mid M) \;+\; \text{SSE}(M)$$

Simplest case: $\widetilde{M}$ is intercept model:

Thus: $\sum_{i=1}^{n} (y_i - \bar{y})^2 = \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$

$\hat{=}$ emp. variance of response, emp. var. of residuals, emp. var. of fitted values

(SST = SSR + SSE)

Deviance analysis

Partitioning of the variation:

variation of Y = variation of the regression + variation of residuals

Corresponding test: F-test
Does a significant increase in the discrepancy exist for the shift from M to $\widetilde{M}$?

$$\frac{(\mathrm{SSE}(\widetilde{M}) - \mathrm{SSE}(M))/r}{\mathrm{SSE}(M))/(n-p-1)} \sim F_{r,n-p-1}$$

Deviance analysis

$\sigma^2 = \phi$

$\sim \sigma^2 \chi^2_r$ if $\tilde{M}$ true

Structure of distribution: Normal distribution, linear model

$$\mathrm{SSE}(\widetilde{M}) \;=\; \mathrm{SSE}(\widetilde{M} \mid M) \;+\; \mathrm{SSE}(M)$$

$\sim \sigma^2 \chi^2_{n-p-1+r}$ if $\tilde{M}$ true

$\sim \sigma^2 \chi^2_{n-p-1}$ if M true

General GLM:

$$D(\widetilde{M}) \;=\; D(\widetilde{M} \mid M) \;+\; D(M)$$

$\stackrel{a}{\sim} \phi \chi^2_{G-p-1+r}$ if $\tilde{M}$ true & grouped data

$\stackrel{a}{\sim} \phi \chi^2_r$ if $\tilde{M}$ true data are grouped or ungrouped

$\stackrel{a}{\sim} \phi \chi^2_{G-p-1}$ if M true & grouped data

Deviance analysis

General model hierarchies

$$M_1 \subset M_2 \subset \ldots \subset M_m$$

$$\begin{aligned}
\mathrm{LR}(M_1, M_m) &= \frac{\sup L(M_1)}{\sup L(M_m)} \\
&= \frac{\sup L(M_1)}{\sup L(M_2)} \cdot \frac{\sup L(M_2)}{\cdots} \cdot \cdots \\
&\quad \cdots \cdot \frac{\cdots}{\sup L(M_{m-1})} \cdot \frac{\sup L(M_{m-1})}{\sup L(M_m)}
\end{aligned}$$

Deviance analysis

Analogous:

$$
\begin{aligned}
D(M_1 \mid M_m) &= D(M_1) - D(M_m) \\
&= \left.D(M_1) - \cancel{D(M_2)}\right\} D(M_1 \mid M_2) \\
&+ \left.\cancel{D(M_2)} - D(M_3)\right\} D(M_2 \mid M_3) \\
&+ \;\ldots. \\
&+ \left.D(M_{m-1}) - D(M_m)\right\} D(M_{m-1} \mid M_m)
\end{aligned}
$$

→ test sequence can be found

$$
\rightarrow \underset{\substack{\text{discrepance}\\ \text{in } M_1}}{D(M_1)} = \underbrace{D(M_1|M_2) + D(M_2|M_3) + \ldots + D(M_{m-1}|M_m)}_{\text{sequential conditional discrepance}} + \underbrace{D(M_m)}_{\text{discrepance of } M_m}
$$

Deviance analysis

Example: with two covariates A and B

Possible models:

M_0 only intercept

M_A intercept + A

M_B intercept + B

M_{AB} : intercept + A + B

M_{AB*} intercept + A + B + Interaction of A and B

$$M_0 \subset M_A \subset M_{AB} \subset M_{AB*}$$

stepwisely compare